

CS198 Project Proposal

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I. Project Title

Explainable AI for Post-COVID Psychological Profile Prediction

II. Project Implementor

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IV. Significance of the Project

A systematic review of clinical databases conducted by Engelmann et al. (2024) found cross-sectional associations between persistent somatic symptoms, particularly Long COVID, and psychological variables such as anxiety and depression. However, they noted a scarcity of studies on Long COVID that explore psychological constructs beyond depression and anxiety, as evidenced by the limited representation of other forms of psychopathology in the reviewed clinical records.

Additionally, a study by Rius Ottenheim et al. (2022) examined how sociodemographic, clinical, and COVID-19-related factors affect changes in patient mental health. Predictor variables, such as sociodemographic and clinical characteristics, influence a person's resilience or vulnerability in stressful situations. Additionally, COVID-19-related factors like changes in activity, teleworking, and caring for children at home were linked to mental health outcomes. They measured these changes using the Quick Inventory of Depressive Symptoms (QIDS) for depression, the Beck Anxiety Inventory (BAI) for anxiety, the Penn State Worry Questionnaire (PSWQ) for worry, and the De Jong Giervald Loneliness Scale for loneliness.

For this study, we refer to a dataset from an online survey conducted on the Japanese population when the number of COVID-19-positive cases was high and when the COVID-19 pandemic was officially declared to be over. The dataset has 92 attributes, including socio-demographic, COVID-19-related, mental health, and coping behavior information from 2,659 respondents (Nuñez, 2023).

After conducting a Wilcoxon signed-rank statistical test on the following mental health metrics: K6 for emotional distress, PHQ-9 for depression, GAD-7 for anxiety, SSS-8 for somatic symptom burden, and the UCLA loneliness scale. An in-depth report of this is attached in Appendix A.

For these tests, we set our null hypothesis as: "There is no significant difference in these mental health metrics during the COVID-19 pandemic and after it." Our alternative hypothesis (for the one-sided test) was: "There is a statistically significant increase in these mental health metrics after COVID-19."

After conducting the Wilcoxon signed-rank test, we found that each metric had a p-value of less than 0.05, allowing us to reject the null hypothesis and infer that there is a statistically significant increase in these mental health metrics after COVID-19—which is concerning.

This leads us to believe that an AI-assisted psychotherapeutic approach may assist in lowering costs and providing better and faster classification of ex-COVID-19 patients according to high, medium, or low risk of chronic psychopathology. Classification of risk using AI-based prediction methods, supplemented with face-to-face risk assessments as necessary, could facilitate matching individuals with the most appropriate preventive psychotherapeutic interventions, especially in resource-limited settings (Ćosić et al., 2021).

However, Ćosić et al. (2021) highlights that methods based on supervised machine learning (ML), while maximizing classification accuracy, often sacrifice model explainability. Explainability is a key component for reliable and trustworthy predictions, personalized treatments, and appropriate preventive suggestions for mental health deterioration in ex-COVID-19 patients.

AI systems typically operate as "black boxes," especially deep learning algorithms. Despite the excellent performance of these models, the decision-making process is often opaque. This leads to efforts in building explainable AI as a prerequisite for deploying critical applications (Singh et al., 2024).

To apply explainable AI (XAI) extensively, we turn to counterfactual explanations to provide more understandable insights (Wachter et al., 2017). Probabilities or statistical relationships alone do not provide satisfying explanations for why events occur. For explanations to be meaningful, they must be selectively chosen. People tend to focus on one or two explanations from an otherwise infinite set of possible causes. Hence, we use contrastive explanations, responding to counterfactual cases, to gain contextualized insights (Miller, 2018).

Counterfactual explanations typically take the form: “If X had not occurred, then Y would not have occurred.” For example: "If I hadn’t taken a sip of this hot coffee, then I wouldn’t have burned my tongue" (Molnar, n.d.).

V. Project Objectives

This brings us to our project objective: to pivot from black-box models and to turn towards the development of explainable models that (1) are transparent, allowing us to see how each factor in the dataset contributes to the resulting classification, (2) provide interpretable classification, and (3) offer contextualized explanations using counterfactual methods (Singh et al., 2024).

We will then aim to package the resulting XAI in a website for people to interact with, and if possible within the timeframe, ground it into a more local setting by using Philippine datasets to train our models.

VI. Review of Related Literature

SHapley Additive exPlanations (SHAP) utilizes game theory to explain the results of black-box models. It is based on classical Shapley value estimation, which is a regression method that computes the weighted average of all possible differences. By computing SHAP values, we are able to point out to each feature the change in the expected model prediction when conditioning on those features (Lundberg & Lee, 2017).

Since the computation of exact SHAP values are costly, especially in large aggregates, we use approximations. For this study, we propose utilizing the Shapley kernel to allow us to start from a broad, model-agnostic solution (Lundberg & Lee, 2017).

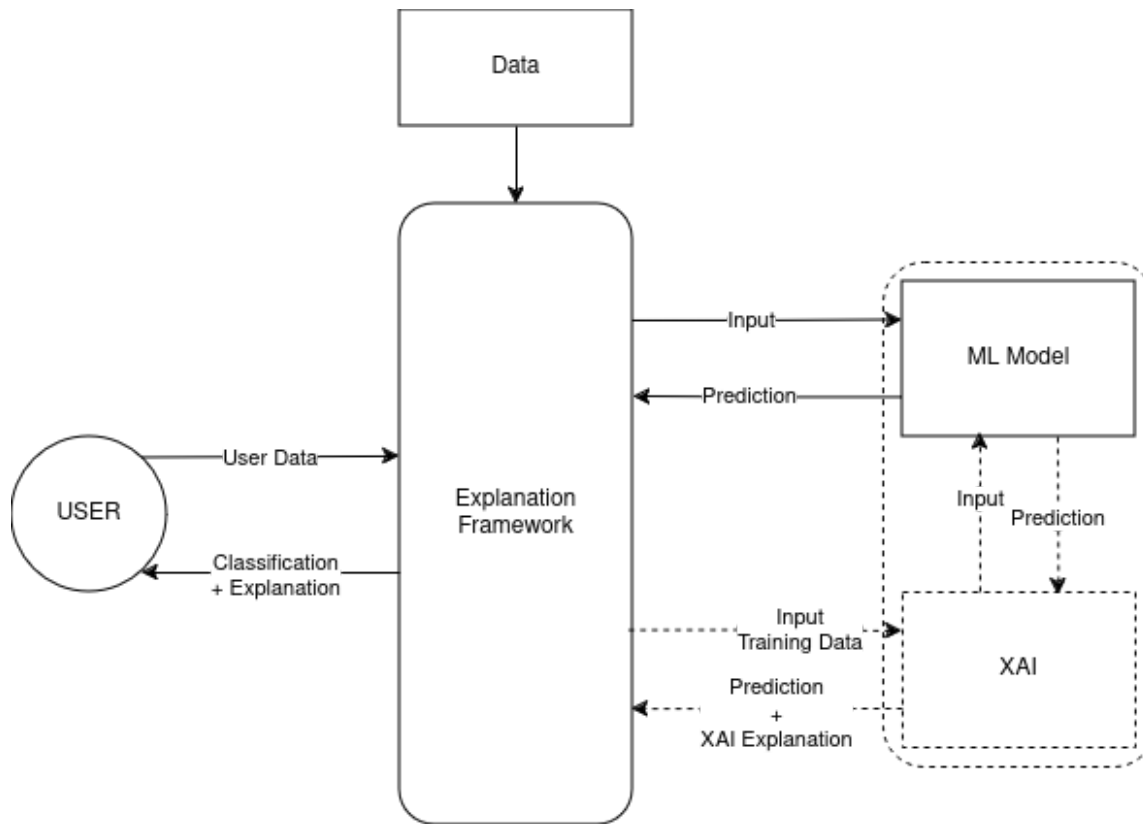
Other explainable method include **Local Interpretable Model-agnostic Explanations (LIME)** which generates local approximations of model predictions (Ribeiro et al., 2016), and **Deep Learning Important Features (DeepLIFT)** which decomposes the output prediction of a neural network by backpropagating contributions of all neurons in the network to each input feature (Shrikumar et al., 2019).

InterpretML is an open-source Python toolkit for explaining black-box AI systems and training intelligible models. It incorporates state-of-the-art machine learning intelligibility techniques developed by Microsoft or third-party libraries. It creates a common API across integrated libraries, making it possible to call and compare different intelligibility methods using the same set of function calls and data structures (Microsoft, 2020). InterpretML also includes **Diverse Counterfactual Explanations (DiCE)**, a package for implementing counterfactual explanations (Mohtilal et al., 2020).

Additionally, Counterfactual SHAP (CF-SHAP) is a modular framework that leverages the fitness of mathematical properties of Shapley values, and by extension SHAP values, with human intuition for feature importance in order to generate counterfactual explanations (Albini et al., 2022).

VII. Theoretical Framework

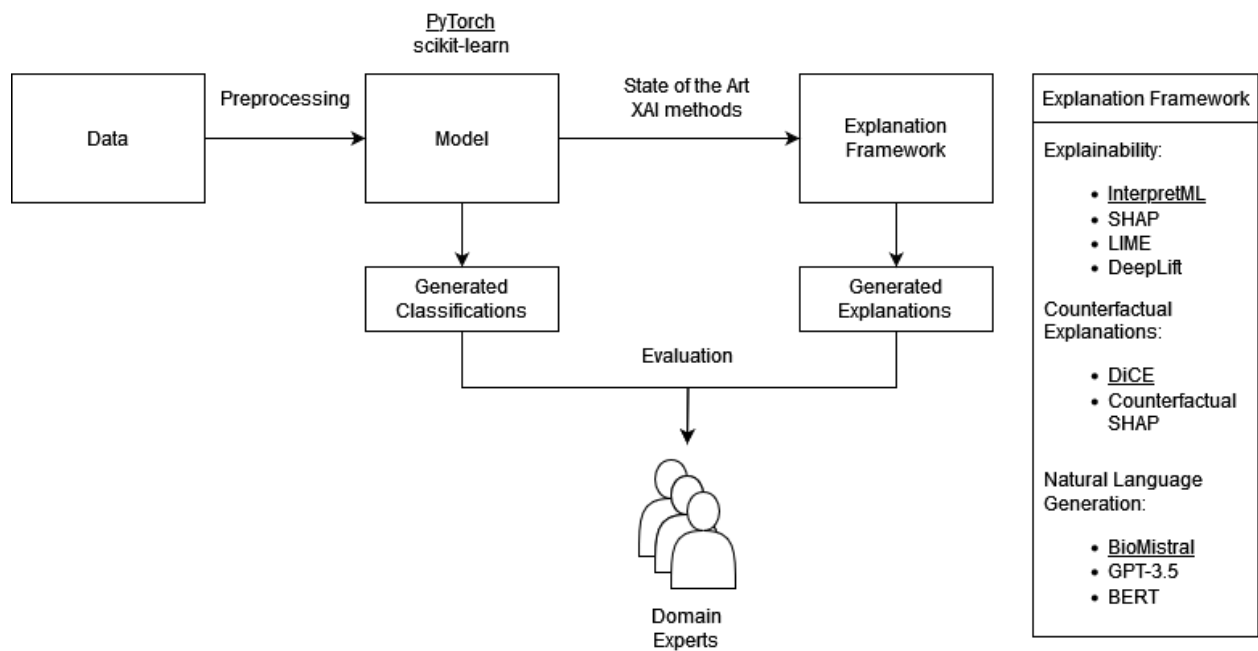
We will adapt Kshetry & Kantardzic's (2024) framework on building XAI that leverages counterfactuals to build contextualized and interpretable explanations for casual understanding.



This focuses on the robust explanation framework that utilizes user data and survey data in order to generate interpretable classifications and contextual explanations. A part of the survey data is used to train the ML Model which provides predictions that will be interpreted by the explanation framework. Similarly the same predictions will be used as a baseline to determine if the XAI is of similar accuracy to the ML Model. The XAI will utilize another part of the survey data to generate a simpler yet similar model to the original one. The XAI will utilize the state of the art intelligible frameworks mentioned in the Review of Related Literature section.

We would also like to add that in order to provide the user with meaningful explanations, natural language models will be explored and utilized in order to synthesize the outputs of the XAI model.

VIII. Proposed Methodology



Our current methodology focuses on the development of the baseline model and the explanation framework.

We begin with preprocessing the data to build our baseline model using PyTorch. We will use the sociodemographic, COVID-19-related, mental health, and coping behavior information columns from our dataset to predict our target features which are the mental health metrics. We will then utilize state-of-the-art XAI methods to implement our Explanation Framework. Generated results from our baseline model and explanation framework will be evaluated by domain experts in the field of psychology. In building our explanation framework, we will utilize InterpretML and DiCE for explainability and counterfactual explanations, alongside BioMistral, which will aid in packaging our counterfactual explanations.

IX. Major Activities

- Data preprocessing and visualization
- Baseline classifier
- Explanation framework
- Model improvement
- Web platform (visualization)

X. Resources Needed

This project would only require domain experts within the field of psychology in order to properly assess the generated classifications and explanations of the model.

XI. Timeline (1st and 2nd sem)

The more detailed Gantt chart could be accessed through this link:

<https://docs.google.com/spreadsheets/d/1yu404o-QjMFbbBwXKJAzHhMj7wzOxZ7g7w0fjb7MndE/edit?gid=1061435246#gid=1061435246>

Activity	Start	End
Identify K6, PHQ9, UCLA thresholds	Sep 23, 2024	Oct 7, 2024
Track K6, PHQ9, UCLA of respondents	Sep 23, 2024	Oct 7, 2024
Pre-process new data	Oct 7, 2024	Oct 21, 2024
Data visualization (animation)	Oct 21, 2024	Nov 11, 2024
Test baseline classifier	Oct 21, 2024	Nov 11, 2024
Test 2 classifiers	Nov 18, 2024	Dec 2, 2024
Data and model visualization	Feb 3, 2025	Feb 24, 2025
Model improvement	Feb 24, 2025	Mar 10, 2025
Web platform (visualization)	Feb 3, 2025	May 19, 2025

XII. Appendix

Appendix A.

K6: 0 to 24 (Kessler et al., 2010)

0-12: Unlikely distress

13-24: Likely distress

Null Hypothesis: There is no significant difference in K6 scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in K6 scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect K6 scores to be generally higher after COVID-19
 - alternative='less': We expect K6 scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=575312.0, pvalue=0.007354128804346646)
The test yielded a test statistic of 575312.0 and a p-value of 0.007354128804346646. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in K6 scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=672098.0, pvalue=0.003677064402173323)
The test yielded a test statistic of 672098.0 and a p-value of 0.003677064402173323. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in K6 scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=672098.0, pvalue=0.9963229355978267)
The test yielded a test statistic of 672098.0 and a p-value of 0.9963229355978267. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant difference in K6 scores during the COVID-19 pandemic and after it.

PHQ-9: (Kroenke et al., 2001)

0-4: Minimal depression

5-9: Mild

10-14: Moderate

15-19: Moderately severe

20-27: Severe

Null Hypothesis: There is no significant difference in PHQ-9 scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in PHQ-9 scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect PHQ-9 scores to be generally higher after COVID-19
 - alternative='less': We expect PHQ-9 scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=620439.0, pvalue=0.0009218442396374339)
The test yielded a test statistic of 620439.0 and a p-value of 0.0009218442396374339. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in PHQ-9 scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=748246.0, pvalue=0.00046092211981871696)
The test yielded a test statistic of 748246.0 and a p-value of 0.00046092211981871696. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in PHQ-9 scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=748246.0, pvalue=0.9995390778801813)
The test yielded a test statistic of 748246.0 and a p-value of 0.9995390778801813. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant decrease in PHQ-9 scores during the COVID-19 pandemic and after it.

GAD-7: (Generalised Anxiety Disorder Assessment (GAD-7), n.d.)

0-4: Minimal anxiety

5-9: Mild

10-14: Moderate

15-21: Severe

Null Hypothesis: There is no significant difference in GAD-7 scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in GAD-7 scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect GAD-7 scores to be generally higher after COVID-19
 - alternative='less': We expect GAD-7 scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=447439.0, pvalue=3.543452134436497e-07)
The test yielded a test statistic of 447439.0 and a p-value of 3.543452134436497e-07. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in GAD-7 scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=608892.0, pvalue=1.7717260672182485e-07)
The test yielded a test statistic of 608892.0 and a p-value of 1.7717260672182485e-07. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in GAD-7 scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=608892.0, pvalue=0.9999998228273933)
The test yielded a test statistic of 608892.0 and a p-value of 0.9999998228273933. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant decrease in GAD-7 scores during the COVID-19 pandemic and after it.

SSS-8: (Gierk et al., 2014)

0-3: Minimal somatic symptom burden

4-7: Low

8-11: Medium

12-15: High

16-32: Very high

Null Hypothesis: There is no significant difference in SSS-8 scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in SSS-8 scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect SSS-8 scores to be generally higher after COVID-19
 - alternative='less': We expect SSS-8 scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=962330.5, pvalue=0.025374664112590598)
The test yielded a test statistic of 962330.5 and a p-value of 0.025374664112590598. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in SSS-8 scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=1078879.5, pvalue=0.012687332056295299)
The test yielded a test statistic of 1078879.5 and a p-value of 0.012687332056295299. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in SSS-8 scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=1078879.5, pvalue=0.9873126679437046)
The test yielded a test statistic of 1078879.5 and a p-value of 0.9873126679437046. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant decrease in SSS-8 scores during the COVID-19 pandemic and after it.

LSNS-6: 0 to 60 (Lubben, 2013)

0 -> Minimal social engagement

60 -> High social engagement

Null Hypothesis: There is no significant difference in LSNS-6 scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in LSNS-6 scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect LSNS-6 scores to be generally higher after COVID-19
 - alternative='less': We expect LSNS-6 scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=1094657.5, pvalue=0.012384572791135867)
The test yielded a test statistic of 1094657.5 and a p-value of 0.012384572791135867. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in LSNS-6 scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=1239222.5, pvalue=0.006192286395567933)
The test yielded a test statistic of 1239222.5 and a p-value of 0.006192286395567933. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in LSNS-6 scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=1239222.5, pvalue=0.9938077136044321)
The test yielded a test statistic of 1239222.5 and a p-value of 0.9938077136044321. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant decrease in LSNS-6 scores during the COVID-19 pandemic and after it.

UCLA Loneliness Scale (UCLA LS): (Deckx et al., 2014)

0-19: Minimal loneliness

20-34: Low

35-49: Moderate

50-64: Moderately high

65-80: High

Null Hypothesis: There is no significant difference in UCLA LS scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in UCLA LS scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect UCLA LS scores to be generally higher after COVID-19
 - alternative='less': We expect UCLA LS scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=1105938.0, pvalue=4.030348402741578e-08)
The test yielded a test statistic of 1105938.0 and a p-value of 4.030348402741578e-08. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in UCLA LS scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=1444473.0, pvalue=2.015174201370789e-08)
The test yielded a test statistic of 1444473.0 and a p-value of 2.015174201370789e-08. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant increase in UCLA LS scores after the COVID-19 pandemic.

One-sided -> less
WilcoxonResult(statistic=1444473.0, pvalue=0.999999979848258)
The test yielded a test statistic of 1444473.0 and a p-value of 0.999999979848258. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant decrease in UCLA LS scores during the COVID-19 pandemic and after it.

SHS (Subjective Happiness Scale): (De Leon et al., 2022)

1 -> Low happiness

7 -> High happiness

Null Hypothesis: There is no significant difference in SHS scores during the COVID-19 pandemic and after it.

Alternative Hypothesis:

- Defining the two-sided test:
 - There is a significant difference in SHS scores during the COVID-19 pandemic and after it.
- Defining the one-sided tests:
 - alternative='greater': We expect SHS scores to be generally higher after COVID-19
 - alternative='less': We expect SHS scores to be generally less after COVID-19

Running the tests:

Two-sided
WilcoxonResult(statistic=1019415.0, pvalue=4.363899830080072e-07)
The test yielded a test statistic of 1019415.0 and a p-value of 4.363899830080072e-07. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant difference in SHS scores during the COVID-19 pandemic and after it.

One-sided -> greater
WilcoxonResult(statistic=1019415.0, pvalue=0.9999997818050085)
The test yielded a test statistic of 1019415.0 and a p-value of 0.9999997818050085. Since the p-value is above the 0.05 threshold, we accept the null hypothesis, indicating no significant increase in SHS scores during the COVID-19 pandemic and after it.

One-sided -> less
WilcoxonResult(statistic=1019415.0, pvalue=2.181949915040036e-07)
The test yielded a test statistic of 1019415.0 and a p-value of 2.181949915040036e-07. Since the p-value is below the 0.05 threshold, we reject the null hypothesis, indicating a statistically significant decrease in SHS scores after the COVID-19 pandemic.

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